

Forces from Channels: Event-Density’s Matter Sector is a Lattice Gauge Theory — Gauge Groups from Multiplicity, Parity Violation from the Non-Abelian Sector, and Chirality from the Arrow

Allen Proxmire

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Preamble: What This Paper Does NOT Claim

1. **The 13 substrate primitives are not derived** (position paper, Paper_087).
 2. **The Standard-Model gauge group $U(1) \times SU(2) \times SU(3)$ is not derived.** This paper derives that gauge groups are $SU(N)$ from channel multiplicity, and that the SM groups *correspond* to multiplicities $\{1, 2, 3\}$. It does **not** derive *why* the stable multiplicities stop at 3 (the uniqueness question, §7).
 3. **The weak force’s specific chirality is not derived.** The paper shows parity violation is confined to the non-abelian sector and that all chirality originates in the arrow’s first commitment; it does **not** derive why the *weak* $SU(2)$ in particular couples to one handedness (the matter-assignment, §7.2), which is entangled with the Standard Model’s own open problems (the strong-CP problem, the CKM phase).
 4. **The chirality-origin result (§6) is an account, not a closed proof.** “All chirality traces to the arrow’s first commitment” is built on established pieces (the baryogenesis single-template lock, the topological quantization of handedness, the time-arrow→spatial-twist “screw”) plus one new synthesis; *which* handedness is selected is a contingent initial condition, as in any spontaneous symmetry breaking.
 5. **No Yang–Mills action, no electroweak/Higgs sector, and no anomaly cancellation are derived** (all deferred, §7). The paper builds the gauge *connection and curvature* — the kinematics — not the field dynamics.
 6. **“P05-transport re-routes channels unitarily” is a structural reading of the polarity-transport primitive** (P05 composition + P04 bandwidth conservation + invertibility between commitments), defensible but not a closed substrate proof.
 7. **Masses, mixing angles, and coupling constants are inherited, not derived;** so are the multiplicities $\{1, 2, 3\}$ themselves.
 8. **No claim that ED is the only consistent substrate ontology.**
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Abstract

Event Density’s substrate is a participation graph on which chains propagate along **channels** (P07), each carrying a $U(1)$ **polarity** phase (P09), transported by the connection primitive **P05**. This paper asks what gauge structure that substrate carries, and answers with three structural results and one unifying account. **(1) The gauge group is the channel count.** A rule-type family of N indistinguishable channels, under bandwidth conservation (P04), has structure group $U(N) = SU(N) \times U(1)$ — so non-abelian $SU(N)$ is

forced by channel multiplicity (P08), grounding what the prior gauge-vocabulary paper (T17) obtained only by analogy; the Standard-Model groups correspond to multiplicities $\{1, 2, 3\}$. **(2) The substrate is a lattice gauge theory.** P05-transport of the N channels — bandwidth-conserving, and invertible between commitments — is a $U(N)$ link variable; the gauge field is the per-edge generator, the field strength is the plaquette holonomy. ED’s matter sector is a non-abelian lattice gauge theory on the *relational* graph (no Brillouin torus) carrying the *retarded* arrow — exactly the structure that escapes the Nielsen–Ninomiya doubling no-go that powers the “the universe cannot be discrete” objection. **(3) Parity violation is a non-abelian phenomenon.** The abelian (single-channel) coupling is chirality-blind — vector, parity-conserving (electromagnetism); only the non-abelian (cross-channel) coupling is chirality-sensitive and can violate parity. ED therefore *forbids* a parity-violating abelian force — a falsifiable prediction, since chiral abelian $U(1)$ s are allowed in general gauge theory. The vector/chiral split of the forces is the single-channel/cross-channel split of the substrate’s two propagation kernels. **(4) All chirality originates in the arrow.** A commitment is a handed event — it carries both a P09 phase and a channel-topology “screw” orientation — and the universe’s *first* commitment imprints both a matter/antimatter reference (the phase) and a parity reference (the screw), locked globally and made maximal by their quantized, topological character. Parity violation and the matter/antimatter asymmetry are then two attributes of one first-arrival imprint, correlated but distinct, reproducing C and P violated separately with CP only partial — unifying the chiral-gauge structure with baryogenesis under the substrate’s signature commitment, the arrow. The paper does **not** derive the SM gauge group, the weak force’s specific chirality, or the Yang–Mills dynamics; the chirality result is an account, not a proof. What ED’s *structure* forces — gauge group from multiplicity, lattice-gauge-theory form, parity-violation-is-non-abelian — is independent of those open quantities.

1. Introduction

1.1 What this paper does

The prior gauge paper in this program (T17, Paper_015) was explicit that it performed a *substrate-vocabulary rewrite* of fiber-bundle gauge theory: it *named* channels the fiber, P05 the connection, and $U(1)$ (P09) the structure group, and obtained non-abelian structure only “by substrate-level analogy.” This paper derives what T17 named. In order of decreasing solidity:

1. **The gauge group is $SU(N)$ from channel multiplicity** (§3) — derived from bandwidth conservation on the N -channel amplitude.
2. **The substrate is a lattice gauge theory** (§4) — P05-transport is a $U(N)$ link variable; on the relational graph with the arrow, it escapes Nielsen–Ninomiya.
3. **Parity violation is confined to the non-abelian sector** (§5) — the abelian force is necessarily vector.
4. **All chirality originates in the arrow’s first commitment** (§6) — an account unifying parity violation with the matter/antimatter asymmetry.

The hard structural results (1–3) lead; the chirality account (4) closes. The frontier — why the multiplicities stop at $\{1, 2, 3\}$, and why the weak force in particular is chiral — is left open and honestly marked (§7).

1.2 Why this matters

The sharpest objection to any discrete-substrate program is the **Nielsen–Ninomiya theorem**: on a regular lattice, chiral fermions are forbidden — they come in mirror-paired doublers that cancel, leaving a parity-symmetric (vector-like) world. Since the real world is chiral (the weak force is parity-violating), “the universe cannot be a discrete lattice” follows. This paper meets the objection head-on. ED *is* discrete, but its matter sector is a lattice gauge theory on a **relational** graph — which has no Brillouin torus — carrying a **retarded arrow** — which makes the transport non-hermitian. Both are exactly the premises Nielsen–Ninomiya requires and ED lacks. The doubling no-go does not bind ED, and chirality is not merely *permitted* but turns out to be the matter-sector fingerprint of the same prithon — the arrow — that the gravity line (Paper GR-II) identified as the source of the gravitational “khronon.” One signature commitment; two

fingerprints.

1.3 How this fits the arc

- **Inherited:** T2 (Paper_103, the unique four-component $\text{Cl}(3,1)$ spinor) and T4 (Paper_106, the Dirac equation form); Paper_090 (the V5 cross-chain kernel — its gauge-compatibility and chirality-sensitivity); the baryogenesis single-template / first-arrival lock (the R4 mechanism).
- **Superseded in part:** T17 (Paper_015) — its postulated “P05 = connection” and “channels = fiber” are *derived* here (§4), and its non-abelian “analogy” is *replaced* by the multiplicity derivation (§3).
- **Downstream / open:** the uniqueness of $\{1, 2, 3\}$, the weak matter-assignment, the Yang–Mills action, electroweak/Higgs, anomalies (§7).

1.4 Conventions and regime of validity

Channels and their multiplicity are substrate-level objects; the gauge fields are stated in the continuum (DCGT) limit. The structural results (1–3) are **multiplicity-independent in form** — they do not depend on the *value* $\{1, 2, 3\}$, only on the existence of finite multiplicity. No strong-coupling, no Higgs sector, and no cosmological claim is made. The chirality account (§6) is a linearized/structural statement about the first commitment, not a dynamical simulation.

2. Primitive Inputs

Substrate primitives (postulated, Paper_087): P02 (participation adjacency), P04 (bandwidth — a conserved, additive, non-negative scalar; **load-bearing**), P05 (polarity-transport along edges — the connection), P07 (channel structure — the directional pathways), P08 (multiplicity — the count of same-rule-type channels), P09 ($U(1)$ -valued polarity — the phase), P11 (commitment irreversibility — the arrow).

Inherited from within the program:

- **T2 (Paper_103):** the unique four-component $\text{Cl}(3,1)$ spinor (the spin double-cover is a Clifford-algebra structure, not a P09 structure).
- **T4 (Paper_106):** the Dirac-equation form (the substrate→Dirac chain’s spinor-construction was left open there; §5.3 here locates it in channel topology).
- **Paper_090 (V5):** the cross-chain correlation kernel — gauge-compatible ($K_{V5} \rightarrow e^{i(\alpha(u_A) - \alpha(u_B))} K_{V5}$), retarded, and (via the baryogenesis coherence-functional analysis) **chirality-sensitive**, in contrast to the single-chain kernel V1, which is chirality-blind.
- **The first-arrival lock (R4):** the baryogenesis result that a continuously-degenerate P09-phase circle is broken at first arrival and then globally enforced by V5 phase-coherence under a no-slack capacity condition.

Inherited mathematical input (standard): $U(N)$ as the group of bandwidth-(norm-)preserving transformations of $\mathbb{C}^N$; lattice-gauge-theory link variables, plaquette holonomy, and the Wilson-loop; the Nielsen–Ninomiya theorem and its premises (a compact Brillouin torus and hermiticity).

Value-layer inputs (inherited): particle masses, mixing angles, and the specific multiplicities $\{1, 2, 3\}$.

3. The Gauge Group is the Channel Count

3.1 The channel bundle

Assemble the substrate into a bundle, grounding T17’s named identifications. The **base** is the emergent spacetime (the coarse-grained loci). The **fiber** at a locus is the channel space — for a given rule-type family, the set of that family’s channels (P07), each carrying a $U(1)$ phase (P09). The **connection** is P05 (§4). The **gauge group** is the structure group: the group of fiber transformations that P05-transport can induce while

preserving substrate structure. The question “which gauge group?” thus reduces to “what is the symmetry of the channel fiber?”

3.2 N indistinguishable channels give $U(N)$

Let a rule-type family have **multiplicity** N (P08) — N channels of the *same* rule-type available at a locus. A chain of that family distributes its participation amplitude over them,

$$\psi = (\psi_1, \dots, \psi_N) \in \mathbb{C}^N,$$

the superposition-over-channels that, before a commitment, is the chain sitting across its available same-type channels. Two substrate facts fix the fiber’s symmetry:

1. **Indistinguishability.** The N channels share a rule-type label, and the rule-type label is the only substrate invariant distinguishing channel families; *within* a family there is no substrate fact distinguishing channel i from channel j . Any relabeling or mixing among them is therefore a symmetry.
2. **Bandwidth conservation** (P04). Bandwidth is a conserved additive scalar, so the total participation $\sum_i |\psi_i|^2$ is preserved.

The transformations of $\mathbb{C}^N$ that mix the components and preserve $\sum_i |\psi_i|^2$ are exactly the **unitary group** $U(N)$, and

$$U(N) = (SU(N) \times U(1)) / \mathbb{Z}_N,$$

where the $U(1)$ factor is the *common* P09 phase (the abelian phase T17 already had) and the $SU(N)$ factor is the *traceless* mixing of the N indistinguishable channels. **The non-abelian gauge group $SU(N)$ is therefore forced by channel multiplicity** — it is the symmetry of N degenerate channels, made unitary by bandwidth conservation. This is the explicit derivation T17 §6.1 reached only “by analogy,” and the multiplicity content P08 flagged as needing treatment.

3.3 The Standard-Model correspondence

Indexed by multiplicity:

multiplicity N	structure group	Standard-Model force
1	$U(1)$	electromagnetism / hypercharge
2	$SU(2) \times U(1)$	weak (doublets)
3	$SU(3) \times U(1)$	strong / color (triplets)

So $U(1) \times SU(2) \times SU(3)$ corresponds to channel-family multiplicities $\{1, 2, 3\}$ — the Standard Model’s singlets, doublets, and triplets. *This is a correspondence, not a derivation:* §3.2 gives $SU(N)$ for any N ; why the stable families are exactly $\{1, 2, 3\}$ is open (§7.1).

4. The Substrate is a Lattice Gauge Theory

4.1 From global symmetry to gauge field

A *global* symmetry is not a force. For $SU(N)$ to be a **gauge** symmetry — with a connection, parallel transport, and field strength — transport must rotate among the N channels *position-dependently*, so that transport around a closed loop need not return to the identity.

4.2 P05-transport is a $U(N)$ link variable

Transport one locus to the next maps the amplitude $\psi(u) \in \mathbb{C}^N$ to $\psi(u') \in \mathbb{C}^N$. Three substrate facts fix the form of that map:

- **Channels re-route through the graph** (P07 composition: channels branch and merge; the N channels at u connect to a different arrangement at u'), so the map mixes components.
- **Bandwidth is conserved** (P04), so the map is an isometry of $\mathbb{C}^N$.
- **Between commitments the map is invertible** — commitment (P11) is the *only* non-invertible event; between commitments the evolution is reversible.

An invertible isometry of $\mathbb{C}^N$ is unitary, so

$$U_{uu'} \in U(N), \quad \psi(u') = U_{uu'} \psi(u).$$

A $U(N)$ matrix on each edge **is** a lattice gauge link variable. P05-transport of multiplicity- N channels is a $U(N)$ lattice connection — not by analogy, but because bandwidth-conserving invertible re-routing of N indistinguishable channels is exactly a unitary link map. (*The structural reading: that P05 re-routes channels rather than acting diagonally is read from P07 composition; it is defensible, not a closed proof.*)

4.3 Gauge field and field strength

- **Gauge field** A : the generator of $U_{uu'} = \exp(iA_{uu'})$, $A \in \mathfrak{u}(N)$. The $SU(N)$ part is the non-abelian gauge field; the $U(1)$ part is the common P09 phase.
- **Field strength** F : the **plaquette holonomy** $U_{\square} = \prod_{\text{loop}} U_{uu'}$. If $U_{\square} \neq 1$, transport around the smallest loop rotates the channel amplitude — non-zero curvature, a genuine gauge field rather than pure gauge; the non-commutativity of $SU(N)$ link variables makes F non-abelian ($F = dA + A \wedge A$).
- **Holonomy** $U_C = \prod_C U_{uu'}$: the substrate realization of the holonomy that the Aharonov–Bohm (Paper_010) and Berry-phase (Paper_009) analyses already use as inherited standard mathematics. This paper grounds it — the connection whose holonomy those papers invoke *is* the P05 re-routing of channels.

4.4 On the right kind of lattice — the Nielsen–Ninomiya escape

ED’s channel sector is therefore a $U(N)$ lattice gauge theory, and two features make it the *right* kind:

- It lives on the **relational participation graph**, not a regular periodic lattice — so there is **no Brillouin torus**. The doubling no-go’s topological engine (the sum of Weyl-point chiralities over the compact torus) has no torus to sum over.
- Transport carries the **retarded arrow** (P05 $\rightarrow$ V1 retardation): the link variables are forward-directed and the structure is **non-hermitian**.

Both are exactly the Nielsen–Ninomiya premises ED lacks (a compact Brillouin torus; hermiticity). So the doubling no-go does not bind ED — the discreteness objection of §1.2 is answered structurally — and the gauge sector ($SU(N)$ from multiplicity) and the fermion sector (the spinor as channel topology, §5.3) are one and the same lattice gauge theory. Between commitments the link variables are unitary; at a commitment (P11) the chain’s distribution over the N channels collapses to one — a non-unitary projection. The gauge field lives in the unitary between-commitment transport; commitments are where the gauge-covariant amplitude is read out.

5. Parity Violation is Non-Abelian

5.1 Two kernels, two chiralities

ED has two propagation kernels. **V1** is the single-chain vacuum kernel (Paper_089); **V5** is the cross-chain correlation kernel (Paper_090). The baryogenesis coherence-functional analysis fixes their chirality behavior: **V1 is chirality-blind** — its propagation is independent of the chain-arrow chirality χ_C ($\partial_{\chi} \mathcal{K}_{V1} = 0$) — while **V5 is chirality-sensitive** — its cross-chain coupling depends on the phase relationship $\chi_C - \chi^*$, because V5 couples to P09 phase *differences* and chirality is itself such a difference. V1 (single-chain) cannot see a phase difference; V5 (cross-chain) is built on them.

5.2 The result

Combine §3–§4 with §5.1:

- The **abelian** force is the single-channel coupling, carried by the chirality-blind V1 $\rightarrow$ the same coupling on both handednesses $\rightarrow$ **vector** $\rightarrow$ **parity-conserving**.
- The **non-abelian** force is the cross-channel coupling, carried by the chirality-sensitive V5 $\rightarrow$ can distinguish handedness $\rightarrow$ **can be chiral** $\rightarrow$ **can violate parity**.

In ED, parity violation can occur only in a non-abelian (cross-channel) force. An abelian force is necessarily vector — it cannot violate parity.

This is a falsifiable structural prediction: general gauge theory *permits* chiral abelian $U(1)$ s; ED forbids them, because the abelian coupling is the chirality-blind V1 kernel. It matches the low-energy world — electromagnetism (abelian) is vector; the weak force (non-abelian) is chiral. The vector/chiral split of the forces is the V1/V5 = single-channel/cross-channel split. V5 is the chirality-bearing kernel *because* it is the cross-channel (non-abelian) one.

5.3 Where the spinor’s handedness lives

This also resolves a puzzle in the substrate $\rightarrow$ Dirac chain (T4, Paper_106 §3.7). The spin double-cover is a $Cl(3,1)$ structure (T2), not a P09 structure: a $U(1)$ phase winds by integers, while $\text{spin-}\frac{1}{2}$ needs the $SL(2, \mathbb{C})$ half-turn. So the spinor’s handedness (γ^5) is realized by **channel topology** (the route P07 §7.4 names for $U(1)/SU(2)/SU(3)$), and **P09 is the vector** $U(1)$ — the phase that dresses the topological skeleton, the electromagnetism-like coupling. This is why a naive “transport route” to chirality comes out vector-like in 3+1 dimensions: transport is the V1 (vector) kernel; the handedness was never going to come from there. It comes from the channel topology, and which handedness is selected is §6.

6. The Arrow is the Origin of Chirality

The structural results (§3–§5) place *where* chirality can live (the non-abelian/V5 sector, the channel topology). This section accounts for *where it comes from*. It is an account built on established pieces plus one synthesis, not a closed proof.

6.1 A commitment is a handed event

The **arrow** (P11 commitment-irreversibility; the strictly-retarded V1) is ED’s only handedness-breaking primitive — it breaks time-reversal. A commitment carries (i) a P09 phase and (ii) a channel-topology orientation. The arrow makes both *directed*: the commitment advances time and, correlated with that advance, twists the P09 phase relative to the channel — a **screw**, whose handedness is a spatial handedness (the mechanism by which a time-handedness becomes a spatial one). With §5.3 (handedness = channel topology), the screw’s sense is the spinor’s chirality.

6.2 The first commitment imprints both references

The baryogenesis lock (R4) establishes that a continuously-degenerate P09-phase circle is broken at first arrival and then globally enforced (V5 phase-coherence forecloses any second admission under the no-slack condition). The **first commitment** — the first instance of the arrow acting — has both a phase and a screw orientation, and if it sets the global reference for each (as R4 has it do for the phase), it sets both at once:

- its **P09 phase** $\rightarrow$ the global χ_C reference $\rightarrow$ the **matter/antimatter** selection (C-type);
- its **channel-topology screw** $\rightarrow$ the global helicity reference $\rightarrow$ the **parity** selection (P-type).

6.3 Correlated but distinct; and maximal

One event, two attributes, two references: the C-lock and the P-lock are *the same first-arrival event* — hence **correlated** — but *different attributes of it* — hence **distinct**. This reproduces the observed pattern: C and P violated *separately*, CP only *partially*. And because handedness is **topological** (a quantized channel-topology class, not a tunable dial), the selection is all-or-nothing: the imprint is **maximal** on both faces — a complete matter/antimatter selection and maximal ($V-A$) parity violation — from one lock.

6.4 The unification

Parity violation and the matter/antimatter asymmetry are two faces of a single event — the first commitment’s handedness imprint — and that imprint is the arrow acting for the first time.

The arrow is thus the source of *all* chirality in ED. In the gravity line (Paper GR-II) the same arrow is the source of the gravitational “khronon,” the preferred-foliation scalar; the arrow also underwrites the position-dependent clock and the density-suppressed preferred-frame parameters. Chirality is its matter-sector fingerprint. One one-way commitment; the khronon in gravity and the handedness of matter in the same stroke.

7. The Residual and the Open Frontier

7.1 Why $\{1, 2, 3\}$? — open

§3 gives $SU(N)$ for any N ; the Standard Model stops at 3. Why the stable channel-families are exactly $\{1, 2, 3\}$ — three forces, of these sizes and no larger — is not derived here. The natural place to look is **channel-family stability**: the multiplicity of a same-rule-type family is “usually small” (P07), bounded by whatever coherence condition keeps N indistinguishable channels mutually stable, and a stability calculation that caps N and populates $\{1, 2, 3\}$ would close the question. The bound, if it exists, is internal — a property of channel-family coherence — and its value is open.

7.2 Why the weak force specifically is chiral — open

§5 confines parity violation to the non-abelian sector, but both the weak ($SU(2)$) and strong ($SU(3)$) forces are non-abelian, and the strong force is *vector*. So chirality is set by the matter **handedness-assignment** — which channel-topology classes a force couples — not by the group itself. Why the weak $SU(2)$ couples to one handedness while the strong $SU(3)$ couples to both is not derived here, and it is entangled with the Standard Model’s own open problems (the strong-CP problem; the CKM-phase origin of CP violation). §6 supplies the global handedness *reference*; it does not single out the weak force as the one that couples to it asymmetrically.

7.3 Deferred: dynamics, electroweak, anomalies

The Yang–Mills *action* (why the gauge-field dynamics are $\propto F^2$) — built from coarse-graining the plaquette structure (lattice $\rightarrow$ continuum via DCGT) — is not derived. The electroweak/Higgs sector is beyond the V1/V5 split (chiral hypercharge and symmetry breaking are not captured by the simple abelian/non-abelian distinction). Anomaly cancellation is untouched, and is the hardest piece.

8. The Wedge — Where ED Predicts

1. **No parity-violating *abelian* fundamental force** (§5.2). Electromagnetism-type (abelian) forces are vector. The sharpest clean prediction — general gauge theory permits chiral abelian forces; ED forbids them.

2. **Gauge-group rank = channel multiplicity** (§3). Fundamental forces come in $SU(N) \times U(1)$ families indexed by an integer multiplicity.
 3. **One lattice gauge theory unifies gauge fields and matter** (§4): the gauge fields (channel multiplicity) and the fermions (channel topology) are the *same* structure on the relational graph — the discreteness objection answered, not evaded.
 4. **Parity violation and the matter/antimatter asymmetry share one origin** (§6): the first-arrival imprint. A *unification* claim — they are not independent.
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9. Falsification Criteria

9.1 A parity-violating abelian force

Falsifier sentence: *Discovery of a fundamental, parity-violating abelian ($U(1)$ -type) gauge force would falsify §5.2 — that the abelian coupling is the chirality-blind single-channel kernel and is therefore necessarily vector.*

9.2 A non-multiplicity gauge group

Falsifier sentence: *A fundamental gauge force whose group is not of the channel-multiplicity form $SU(N) \times U(1)$ — for instance an exceptional-group gauge force not reducible to a multiplicity structure group — would falsify §3.*

9.3 Doubling on the relational lattice

Falsifier sentence: *A demonstration that ED’s relational-graph, retarded-arrow lattice gauge structure nonetheless forces fermion doubling (a vector-like spectrum) would falsify the §4.4 Nielsen–Ninomiya escape and the discreteness reply.*

9.4 Uncorrelated C and P

Falsifier sentence: *A demonstration that the matter/antimatter asymmetry and the parity-violation handedness are independent — not a shared first-arrival imprint — would falsify the §6.4 unification.*

10. Position Statement

What this paper claims. Event-Density’s matter sector is a non-abelian **lattice gauge theory** on the relational participation graph; its gauge groups are $SU(N) \times U(1)$ **from channel multiplicity** (P08) under bandwidth conservation (P04); the relational-graph-plus-retarded-arrow structure **escapes the Nielsen–Ninomiya** doubling no-go, answering the discreteness objection; **parity violation is confined to the non-abelian sector** (the abelian force is necessarily vector, a falsifiable prediction); and **all chirality originates in the arrow’s first commitment**, which imprints both the matter/antimatter and parity references — unifying parity violation with baryogenesis. The structural results are independent of the open quantities.

What this paper does not claim. It does not derive the Standard-Model gauge group (why the multiplicities stop at $\{1, 2, 3\}$ is open); it does not derive the weak force’s specific chirality (the matter handedness-assignment is open); it does not derive the Yang–Mills dynamics, the electroweak/Higgs sector, or anomaly cancellation (deferred); §6 is an account, not a closed proof, and the selection direction is a contingent initial condition (spontaneous symmetry breaking). κ , masses, mixings, and the multiplicities themselves are inherited.

Series context. Paper MS-I of the matter-sector line, companion to the gravity line (Paper GR-II): the arrow’s fingerprint in matter (chirality) as the khronon is its fingerprint in gravity. The remaining work —

the multiplicity uniqueness via channel-family stability, the weak matter-assignment, the Yang–Mills action, the electroweak sector, and anomalies — is declared open (§7).

Appendix: Cross-References and Glossary

Cross-references

- **Position paper:** Paper_087.
- **T2 (Paper_103):** $Cl(3, 1)$ four-component spinor (inherited).
- **T4 (Paper_106):** Dirac equation form (§5.3 locates its open spinor-construction in channel topology).
- **T17 (Paper_015):** gauge-vocabulary rewrite (superseded in part — §3, §4).
- **Paper_090:** V5 cross-chain kernel (gauge-compatible, chirality-sensitive).
- **Paper_089:** V1 single-chain kernel (chirality-blind, the single cone).
- **Paper_010 / Paper_009:** Aharonov–Bohm / Berry-phase holonomy (grounded in §4.3).
- **Paper GR-II:** the gravity line — the arrow’s gravitational fingerprint (the khronon).

Glossary

- **Channel (P07).** A stable, directional, rule-type-selective pathway in the participation graph — “a direction a chain can keep going in.”
- **Multiplicity (P08).** The count of same-rule-type channels in a region; the source of the gauge-group rank N .
- **Polarity / P09.** The $U(1)$ phase carried along a channel — the vector ($U(1)$) gauge coupling (electromagnetism).
- **V1 / V5 kernels.** Single-chain (chirality-blind, abelian, vector/EM) and cross-chain (chirality-sensitive, non-abelian, where parity violation lives).
- **Structure group / link variable.** The $U(N)$ symmetry of N indistinguishable channels (structure group); the per-edge $U(N)$ transport (link variable).
- **Plaquette holonomy.** Transport around the smallest loop; the lattice field strength.
- **Screw.** A commitment’s correlated time-advance and P09-phase twist; the mechanism turning the time-arrow into a spatial handedness.
- **First-arrival imprint.** The first commitment’s setting of the global matter/antimatter (phase) and parity (screw) references; locked by R4, maximal because topological.
- **C-type / P-type chirality.** Matter/antimatter (charge-conjugation, the P09 phase) vs parity (helicity, the channel topology) — two attributes of the one imprint.

Reader map and open work

Where to look next. Spin/Dirac structure: T2/T4. The V5 kernel: Paper_090. The first-arrival lock: the baryogenesis arc (R4). The arrow’s gravitational fingerprint: GR-II. **Open work (declared, §7).** The multiplicity uniqueness $\{1, 2, 3\}$ via channel-family stability; the weak matter-assignment; the Yang–Mills action (lattice $\rightarrow$ continuum); the electroweak/Higgs sector; anomaly cancellation.